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MODULE - 3

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COUNTING TECHNIQUES

There are two main principles that are:

1. Sum Rule

Suppose an event can occur in m ways & a second event can occur in n ways and they can occur simultaneously then E or F occurs $m+n$ ways. note:

Eg:

There are 5 female & 8 male profs handling Calculus subject. A student can choose how many ways?

$$\text{no. of ways} = 5 + 8 = 13.$$

2. Product Rule

Suppose an event E can occur in m ways & F is another event can occur in n ways and it is independent. Then E and F can occur in $m \cdot n$ ways.

Eg:

Chairs of an auditorium are to be labelled with a letter and the integer less than 100. How many unique chairs?

$$\text{no. of chairs} = 26 \times 100 = 2600.$$

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* PIGEON HOLE PRINCIPLE

If there are 'p' pigeons and 'h' holes then $h < p$, such that atleast one hole contains atleast $\lceil \frac{p}{h} \rceil$ pigeons, i.e. minimum number of pigeons.
 ↳ finding lower bound

Note:

$$\lceil \frac{p}{h} \rceil = k$$

$$p = h(k-1) + 1$$

Ex:

1.

Among 100 peoples atleast how many of them celebrating their bday in the same month of particular year?

$$p = 100 ; h = 12$$

$$\text{each month} = \lceil \frac{100}{12} \rceil = 9 \#$$

2.

$$p = 367 \text{ people}$$

$$h = 365 \text{ days of a year}$$

$$\text{same day} = \lceil \frac{367}{365} \rceil = 2 \#$$

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3.

min num of students required in the class, to be secured that atleast 6 students will receive the same grade? If there are 5 grades.

$$P = H(k-1) + 1$$

$$\text{Here, } H = 5 ; K = 6$$

$$P = 5(6-1) + 1 = 25 + 1 \\ = 26 //$$

* PERMUTATION

Any arrangement of a set of n objects in a given order r , is called permutation.

$$P(n, r) = \frac{n!}{(n-r)!} = {}^n P_r \quad (\text{without repetition})$$

$$P(n, r, r_1, r_2, \dots, r_n) = \frac{n!}{r_1! r_2! r_3! \dots} \quad (\text{with repetition})$$

eg:

6 objects, say : a, b, c, d, e, f, taken 3 at a time.

$$\text{no. of ways} = \frac{6!}{(6-3)!} = \frac{6!}{3!} = 6 \times 5 \times 4 = 120 //$$

Ex.

1.

How many 7 letter-words can be formed using the letters of BENZENE.

3 E ; 2 N

$$P(7, 3, 2) = \frac{7!}{3!2!} = \frac{7 \times 6 \times 5 \times 4 \times 2}{2} = 420 //$$

2.

How many 5 letter words can be formed using letters of BABBY?

$$P(5, 3) = \frac{5!}{3!} = 5 \times 4 = 20 //$$

* COMBINATION

A combination of n objects taken r at a time is any selection r of the objects ; where order does not count, then

$$C(n, r) = {}^nC_r = \frac{n!}{(n-r)! r!}$$

Eg.

4 objects : A, B, C, D taken 3 at a time.

$${}^4C_3 = \frac{4!}{3!} = 4 //$$

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Ex.

1.

How many committee of 3 can be formed from 8 people?

$$\text{no. of committees} = {}^8C_3 = \frac{8!}{5!3!} = \frac{8 \times 7 \times 6}{6} = 56 //$$

2.

A farmer buys 3 cows, 2 pigs & 4 hens from a man who has 6 cows, 5 pigs & 8 hens. How many choices does the farmer have?

$$\begin{aligned} \text{no. of choices} &= {}^6C_3 \times {}^5C_2 \times {}^8C_4 \\ &= \frac{6 \times 5 \times 4}{6} \times 10 \times \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2} \\ &= 20 \times 20 \times 35 \\ &= 14000 // \end{aligned}$$

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* PRINCIPLE OF INCLUSION AND EXCLUSION

• Inclusion

$$n(A \cup B) = [n(A) + n(B)] - n(A \cap B).$$

$$n(A \cup B \cup C) = [n(A) + n(B) + n(C)] - [n(A \cap B) + n(B \cap C) + n(A \cap C)] + n(A \cap B \cap C).$$

$$\begin{aligned} n(A \cup B \cup C \cup D) &= [n(A) + n(B) + n(C) + n(D)] - [n(A \cap B) + n(A \cap C) + n(A \cap D) + n(B \cap C) + n(B \cap D) + n(C \cap D)] \\ &\quad + [n(A \cap B \cap C) + n(A \cap B \cap D) + n(A \cap C \cap D) + n(B \cap C \cap D) + n(A \cap B \cap C \cap D)] - n(A \cap B \cap C \cap D) // \end{aligned}$$

Ex.

1. Find no. of integers b/w 1-250 (both inclusive) that are divisible by any of the integers : 2, 3, 5, 7.

let A be set of integers divisible by 2

B " " by 3

C " " by 5

D " " by 7

$$n(A) = \left\lfloor \frac{250}{2} \right\rfloor = 125$$

$$n(ANB) = \left\lfloor \frac{250}{6} \right\rfloor = 41$$

$$n(B) = \left\lfloor \frac{250}{3} \right\rfloor = 83$$

$$n(BNC) = \left\lfloor \frac{250}{15} \right\rfloor = 16$$

$$n(C) = \left\lfloor \frac{250}{5} \right\rfloor = 50$$

$$n(CND) = \left\lfloor \frac{250}{35} \right\rfloor = 7$$

$$n(D) = \left\lfloor \frac{250}{7} \right\rfloor = 35$$

$$n(ANC) = \left\lfloor \frac{250}{10} \right\rfloor = 25$$

$$n(AND) = \left\lfloor \frac{250}{14} \right\rfloor = 17$$

$$n(ANBNC) = \left\lfloor \frac{250}{6 \cdot 5} \right\rfloor = 8$$

$$n(BND) = \left\lfloor \frac{250}{21} \right\rfloor = 11$$

$$n(ANBND) = \left\lfloor \frac{250}{6 \times 7} \right\rfloor = 5$$

$$n(BNCND) = \left\lfloor \frac{250}{15 \times 7} \right\rfloor = 2$$

$$n(ANCND) = \left\lfloor \frac{250}{70} \right\rfloor = 3$$

$$n(ANBNCND) = \left\lfloor \frac{250}{210} \right\rfloor = 1$$

$$\begin{aligned} \text{no. of integers} &= (125 + 83 + 50 + 35) - (41 + 16 + 7 + 25 + 17 + 11) \\ &\quad + (8 + 5 + 2 + 3) - 1 \end{aligned}$$

$$= 193 //$$

NOTE:

Qn can be asked to find not divisible numbers too.
To find: $n(\text{universal}) - n(\text{divisible})$.

2.

Find the no. of the int ≤ 1000 and not divisible by 3, 5, 7, 22.

Let A be set of int div by 3

Let B " 5

Let C " 7

Let D " 22

$$n(A) = \left\lfloor \frac{1000}{3} \right\rfloor = 333 \quad n(A \cap B) = \left\lfloor \frac{1000}{15} \right\rfloor = 66$$

$$n(B) = \left\lfloor \frac{1000}{5} \right\rfloor = 200 \quad n(B \cap C) = \left\lfloor \frac{1000}{35} \right\rfloor = 28$$

$$n(C) = \left\lfloor \frac{1000}{7} \right\rfloor = 142 \quad n(C \cap D) = \left\lfloor \frac{1000}{154} \right\rfloor = 6$$

$$n(D) = \left\lfloor \frac{1000}{22} \right\rfloor = 45 \quad n(A \cap C) = \left\lfloor \frac{1000}{21} \right\rfloor = 47$$

$$n(A \cap D) = \left\lfloor \frac{1000}{66} \right\rfloor = 15 \quad n(A \cap B \cap C) = \left\lfloor \frac{1000}{105} \right\rfloor = 9$$

$$n(B \cap D) = \left\lfloor \frac{1000}{110} \right\rfloor = 9 \quad n(A \cap B \cap D) = \left\lfloor \frac{1000}{330} \right\rfloor = 3$$

$$n(A \cap C \cap D) = \left\lfloor \frac{1000}{462} \right\rfloor = 2 \quad n(B \cap C \cap D) = \left\lfloor \frac{1000}{770} \right\rfloor = 1$$

$$n(A \cap B \cap C \cap D) = \left\lfloor \frac{1000}{1540} \right\rfloor = 0$$

$$\therefore n(A \cup B \cup C \cup D) = 564 //$$

$$\therefore n(\overline{A \cup B \cup C \cup D}) = 1000 - 564 = 436$$

3.

$$n(U) = 2500$$

Let A be course in C
B " Pascal
C " Networking

$$n(A) = 1700 ; n(B) = 1000 ; n(C) = 550$$

$$n(A \cap B) = 750 ; n(B \cap C) = 275 ; n(A \cap C) = 400$$

$$n(A \cap B \cap C) = 200$$

$$(i) n(A \cup B \cup C) = ?$$

$$= n(A) + n(B) + n(C) - [n(A \cap B) + n(B \cap C) + n(A \cap C)] + n(A \cap B \cap C)$$

$$= 1700 + 1000 + 550 - (750 + 275 + 400) + 200$$

$$= 2025 //$$

$$(ii) n(\overline{A \cup B \cup C}) = ?$$

$$= 2500 - 2025$$

$$= 475 //$$

Note:

qn can be asked to find num which are not divisible by 2/3/5 & divisible by 7.

$$\text{find } n(\overline{A \cup B \cup C}) \text{ and to find : } n(\overline{A \cup B \cup C \cup D})$$

$$= \left\lfloor \frac{4}{7} \right\rfloor$$

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* RECURRENCE RELATION

An equation that expresses in terms of one or more no. of preceding terms of the sequence.

eg: $\{a_n\}$, $\{F_n\}$, $\{T_n\}$, $\{P_n\}$, $\{S_n\}$

RE : $S_n = S_{n-1} + 1$; soln : $S_n = n$

Ex.

1.

IC : $S_0 = 1$

$S_1 = 2 = 2 \times S_0$

$S_2 = 4 = 2 \times S_1$

$\therefore$ RE : $S_n = 2 \times S_{n-1}$; soln : $S_n = 2^n$

2.

Fibonacci series

IC : $\begin{cases} S_0 = 1 \\ S_1 = 1 \end{cases}$

$S_2 = 2 = S_0 + S_1$

$S_3 = 3 = S_1 + S_2$

RE : $S_n = S_{n-1} + S_{n-2}$; soln : $f(n)$

3.

RE : $a_n = a_{n-1} + 3$, $n \geq 1$

IC : $a_0 = 2$

$a_1 = a_0 + 3 = 5$

$a_2 = a_1 + 3 = 8$

$a_3 = a_2 + 3 = 11$

4.

$$RE: a_n = a_{n-1} - a_{n-2}, \quad n \geq 2$$

$$IC: a_0 = 3; a_1 = 5$$

$$a_2 = a_1 - a_0 = 2$$

$$a_3 = a_2 - a_1 = -3 //$$

* SOLVING 1st & 2nd ORDER L.R.E

$$\text{Solution of } a_n = a_h + a_p$$

$\downarrow$ $\rightarrow$ particular soln
 homogeneous soln

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} \dots c_k a_{n-k} + f(n)$$

$\rightarrow$ recurrence relation of order k , where $c_1, c_2 \dots$ are real numbers.

if $f(n) = 0$: homogeneous

$f(n) \neq 0$: non-homogeneous

$\rightarrow$ is zero for homogeneous

general soln: $a_n = a_h + \boxed{a_p}$

* FINDING a_h

if roots are:

$$1. m_1 \neq m_2 : a_h = A m_1^n + B m_2^n$$

$$2. m_1 = m_2 = m : a_h = (An + B)m^n$$

* FINDING a_p

if $f(n)$ is:

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1. $(b_0 + b_1 n + b_2 n^2 \dots) s^n$: $a_p = (p_0 + p_1 n + p_2 n^2 \dots p_k n^k)$
 where $b_0, b_1, s \in \mathbb{R}$ & s is not a root.

5. when s is a root : $a_p = (p_0 + p_1 n + p_2 n^2 \dots p_k n^k) n^s$

Ex.

1.

$$a_n = 2a_{n-1} + 3^n$$

$$IC: a_1 = 5$$

$$GS: a_n + a_p$$

$$a_n - 2a_{n-1} = 3^n \rightarrow 1^{st} \text{ order } \textcircled{1}$$

$$\textcircled{1} - 2) \downarrow$$

 $a_n:$

$$m - 2 = 0$$

$m = 2$, one distinct root

$$a_p = A(2)^n$$

 $a_p:$

$f(n) = 3^n$, since 3 is not a root

trial soln: $a_p = A \cdot 3^n$, subst. in $\textcircled{1}$

$$A \cdot 3^n - 2A(3^{n-1}) = 3^n$$

$$3A - 2A = 3$$

$$A = 3_{//}$$

$$\therefore a_p = 3 \cdot 3^n = 3^{n+1}$$

$$\therefore GS: a_n = A \cdot 2^n + 3^{n+1}$$

$a_p = P \cdot n \cdot 5^n = \text{trial soln, substit in}$

①

$$P_n(5^n) - 3P(n-1)(5^{n-1}) - 10P(n-2)(5^{n-2}) = 7 \cdot 5^n$$

$$a_1 = A \cdot 2^1 + 3^2 \quad 25P(n) - 3P(n-1)(5) - 10P(n-2)$$

$$5 = 2A + 9$$

$$2A = -4$$

$$A = -2$$

$$\therefore a_n = -2^{n+1} + 3^{n+1}$$

$$35P = 7 \cdot (25)$$

$$P = 5$$

2.

$$f(n) = 3 \cdot f_{n-1} + 10 \cdot f_{n-2} + 7(5^n) \quad \text{IC: } f_0 = 4$$

$$f_1 = 3$$

$$f_n - 3f_{n-1} - 10f_{n-2} = 7 \cdot 5^n \rightarrow \text{①}$$

Ans:

$$a_p = n 5^{n+1}$$

$$m^2 - 3m - 10 = 0$$

$$m(m-5) + 2(m-5) = 0$$

$$m = 5 / m = -2, \text{ distinct}$$

$$a_n = A \cdot 5^n + B(-2)^n + n 5^{n+1}$$

$$a_0 = A - B = 4$$

$$a_1 = 5A - 2B + 25 = 3$$

$$a_n : A(5)^n + B(-2)^n$$

$$5A - 2B = -22$$

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Ex.

$$a_n + 6a_{n-1} + 9a_{n-2} = (-3)^n \quad n \geq 2$$

$$\hookrightarrow \textcircled{1}$$

$$\text{IC: } a_0 = 2$$

$$a_1 = 3$$

$$\text{soln: } a_n = a_h + a_p$$

 a_h

$$m^2 + 6m + 9 = 0$$

$$m = -3/-3$$

$$a_h = (A_n + B)(-3)^n$$

 a_p

$$\text{let trial soln } a_p = a_n = An^2(-3)^n$$

 $\text{subst in } \textcircled{1}$

$$An^2(-3)^n + 6A(n-1)^2(-3)^{n-1} + 9A(n-2)^2(-3)^{n-2} = (-3)^n$$

$$An^2(9) + 6A(n-1)^2(-3) + 9A(n-2)^2 = 9$$

$$A(9n^2 + (-18)(n-1)^2 + 9(n-2)^2) = 9$$

$$A(n^2 - 2(n^2 + 1 - 2n) + (n^2 + 4 - 4n)) = 1$$

$$A(2) = 1$$

$$\boxed{A = 1/2}$$

$$\therefore a_p = \frac{n^2(-3)^n}{2}$$

$$a_n = (A_n + B)(-3)^n + \frac{(-3)^n n^2}{2}$$

$$a_0 = B = 2.$$

$$a_1 = (A+B)(-3) + \frac{(-3)}{2} = 3$$

$$-6(2+A) - 3 = 6$$

$$-4 - 2A - 1 = 2$$

$$2A = -7$$

$$A = -7/2$$

$$a_n : \left(\frac{-7n}{2} + 2 \right) (-3)^n + \frac{(-3)^n n^2}{2} //$$

* GENERATING FUNCTION

Generating function represents sequences where each term of a sequence is expressed as a coefficient of variable x in a formal power series.

$$\text{eg: } a_0 + a_1 x + a_2 x^2 + \dots = \sum_{n=0}^{\infty} a_n x^n = G(x)$$

$G(x)$ for known sequences:

$$1. a_n = \frac{1}{2^n}, \quad n \geq 0$$

$$G(x) = \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n = \left(1 - x/2\right)^{-1} = \frac{2}{2-x}$$

$$2. a_n = \frac{1}{n!}, \quad n \geq 0$$

$$G(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

$$3. a_n = a^n, \quad n \geq 0$$

$$G(x) = \sum_{n=0}^{\infty} (ax)^n = (1-ax)^{-1} = \frac{1}{1-ax} \quad |ax| < 1$$

$$4. a_n = (n+1), \quad n \geq 0$$

$$G(x) = \sum_{n=0}^{\infty} x^n (n+1) = 1 + 2x + 3x^2 + \dots = \frac{1}{(1-x)^2} = (1-x)^{-2}$$

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Ex.

1.

use generating function, solve:

$$a_n - 2a_{n-1} = 3^{n-1}, \quad n \geq 1 \quad \text{IC: } a_0 = 5$$

$$\Rightarrow a_n \cdot x^n - 2 \cdot a_{n-1} \cdot x^n = 3^{n-1} \cdot x^n$$

$$\Rightarrow \sum_{n=1}^{\infty} a_n \cdot x^n - 2 \sum_{n=1}^{\infty} x^n \cdot a_{n-1} = \sum_{n=1}^{\infty} 3^{n-1} \cdot x^n$$

$$\Rightarrow [a_1 x + a_2 x^2 + a_3 x^3 \dots] - 2 [a_0 x + a_1 x^2 + a_2 x^3 \dots] = [x + 3x^2 + 9x^3 \dots]$$

$$\Rightarrow G(x) - a_0 - 2xG(x) = x(1-3x)^{-1}$$

$$G(x)[1-2x] = \frac{x}{1-3x} + a_0$$

$$G(x) = 5 + \frac{x}{1-3x} = \frac{5-14x}{1-3x} \cdot \frac{1}{(1-2x)}$$

$$\therefore G(x) = \frac{5-14x}{(1-3x)(1-2x)}$$

$$5-14x = A(1-2x) + B(1-3x)$$

$$x = 1/2$$

$$x = 1/3$$

$$5-7 = -2 = B \left(1 - \frac{3}{2}\right)$$

$$5 - \frac{14}{3} = \frac{1}{3} = A \left(1 - \frac{2}{3}\right)$$

$$-4 = -B$$

$$\boxed{B=4}$$

$$\frac{1}{3} = \frac{A}{3}$$

$$\boxed{A=1}$$

$$G(x) = \frac{1}{1-3x} + \frac{4}{1-2x}$$

$$G(x) = (1-3x)^{-1} + 4(1-2x)^{-1} //$$

$$\therefore \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (3x)^n + 4 \sum_{n=0}^{\infty} (2x)^n //$$

$$\therefore a_n = 3^n + 2^{n+2}, \quad n \geq 0 //$$

2.

$$a_n = 2a_{n-1} + 4a_{n-2}, \quad n \geq 2 \quad \text{IC: } a_0 = 1$$

$$a_1 = 3$$

$$a_n \cdot x^n = 2 \cdot a_{n-1} \cdot x^n + 4 \cdot a_{n-2} \cdot x^n$$

$$\sum_{n=0}^{\infty} a_n \cdot x^n = 2 \sum_{n=2}^{\infty} a_{n-1} x^n + 4 \sum_{n=2}^{\infty} a_{n-2} \cdot x^n$$

$$[a_2 x^2 + a_3 x^3 \dots] = 2[a_1 x^2 + a_2 x^3 \dots] + 4[a_0 x^2 + a_1 x^3 \dots]$$

$$-(a_0 + a_1 x) + G(x) = 2x G(x) - 2a_0 x + 4x^2 \cdot G(x)$$

$$G(x) [1 - 2x - 4x^2] = -2a_0 x + a_0 + a_1 x$$

$$= -2x + 1 + 3x$$

$$-G(x)(4x^2 + 2x + 1) = x + 1$$

$$G(x) = \frac{-(x+1)}{(4x^2+2x+1)} = \frac{Ax+B}{4x^2+2x-1}$$

$$A = -1, B = -1 //$$

This is an improper fraction //

3.

$$a_n = 9a_{n-1} + (10)^{n-1}, \quad n \geq 1 \quad \text{IC: } a_0 = 9$$

$$a_n \cdot x^n = 9a_{n-1} \cdot x^n + 10^{n-1} \cdot x^n$$

$$\sum_{n=1}^{\infty} a_n x^n - 9 \sum_{n=1}^{\infty} a_{n-1} \cdot x^n = \sum_{n=1}^{\infty} 10^{n-1} \cdot x^n = (0)$$

$$[a_1 x + a_2 x^2 \dots] - 9[a_0 x + a_1 x^2 \dots] = [x + 10x^2 + 100x^3 \dots]$$

we know, $g(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 \dots$

$$g(x) - a_0 - 9x \cdot g(x) = x \cdot (1-10x)^{-1}$$

$$g(x) [1-9x] = \frac{x}{1-10x} + 9$$

$$= \frac{9-89x}{1-10x}$$

$$g(x) = \frac{9-89x}{(1-10x)(1-9x)} //$$

$$9-89x = A(1-9x) + B(1-10x)$$

$$x = 1/9$$

$$9 - \frac{89}{9} = \frac{-8}{9} = B \left(1 - \frac{10}{9}\right)$$

$$\frac{-8}{9} = \frac{-B}{9} \Rightarrow \boxed{B=8}$$

$$x = 1/10$$

$$9 - \frac{89}{10} = \frac{1}{10} = A \left(1 - \frac{9}{10}\right)$$

$$\frac{1}{10} = \frac{A}{10} \Rightarrow \boxed{A=1}$$

$$g(x) = \frac{1}{1-10x} + \frac{8}{1-9x}$$

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$$G(x) = (1-10x)^{-1} + 8 \cdot (1-9x)^{-1}$$

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (10x)^n + 8 \sum_{n=0}^{\infty} (9x)^n$$

$$\therefore a_n = 10^n + 8 \cdot 9^n //$$

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Ex.

$$a_n - 6a_{n-1} + 9a_{n-2} = 0, n \geq 2$$

$$IC: a_0 = 1; a_1 = 9$$

$$a_n \cdot x^n - 6a_{n-1} \cdot x^n + 9 \cdot a_{n-2} \cdot x^n = 0$$

$$\sum_{n=2}^{\infty} a_n \cdot x^n - \sum_{n=2}^{\infty} 6 \cdot a_{n-1} \cdot x^n + 9 \sum_{n=2}^{\infty} a_{n-2} \cdot x^n = 0$$

$$[a_2 x^2 + a_3 x^3 \dots] - 6[a_1 x^2 + a_2 x^3 \dots] + 9[a_0 x^2 + a_1 x^3 \dots] = 0$$

$$G(x) - a_0 - a_1 x - 6G(x) \cdot x - 6a_0 x + 9x^2 \cdot G(x) = 0$$

$$G(x) [1 - 6x + 9x^2] - 1 - 9x - 6x = 0$$

$$G(x) = \frac{1+15x}{1-6x+9x^2}$$

$$= \frac{1+15x}{(1-3x)^2}$$

$$1+15x = A(1-3x) + B$$

$$G(x) = \frac{-5}{1-3x} + \frac{6}{(1-3x)^2}$$

$$x = 1/3$$

$$x = 0$$

$$= -5(1-3x)^{-1} + 6(1-3x)^{-2}$$

$$6 = B$$

$$1 = 6 + A$$

$$A = -5$$

$$G(x) = -5 \sum_{n=0}^{\infty} (3x)^n + 6 \sum_{n=0}^{\infty} x^n (n+1)$$

Not x^n it should be $(3x)^n$

$$G(x) = \sum_{n=0}^{\infty} a_n \cdot x^n$$

$$\therefore a_n = -5 \cdot 3^n + 6 \cdot (n+1)$$

Combin